BADG 2: A Perfect Game

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Abstract

An earlier (b. a. d. g.) paper (http://fettermania.com/math/bdice.pdf) focused on a near-optimal strategy for maximizing expected value in gameplay. This paper focuses on a different question: what is the probability of playing a perfect game? Using standard recursive calculation, we calculate the probabilities with the out-of-box game setup. Then we venture into a related question: What's the expected win probability with n standard dice of side count s as $n \to \infty$? We give poor bounds, prove convergence, and find surprising results along the way.

1 Introduction and the Commercial Game

Outside of ideological patron Larry Waldman, the publication of my original b. a. d. g. paper (http://fettermania.com/math/bdice.pdf) received little fanfare from humans. However, robots.txt somehow picked up on this and brought this to the attention of Samuel Cox, creator of the game. After dispensing mild praise about the paper and strong confusion about my interest in it, Mr. Cox suggested another problem.

Instead of "how does one play the game with high/maximum expected value?" (addressed in the first paper), Mr. Cox wondered how often a *perfect game* (one in which the best possible score is achieved at the end) occurs? There's a computational aspect to this, but an implied question that Mr. Cox has debated with his class is also: "is it always best to pick up all your maximum-roll dice?"

I took this on and a day or so later, the solution to this particular question surfaced readily. Naturally unsatisfied by instances, I sought a more general solution to a subproblem I found. Nine months later, just as I was beginning to enjoy my futility and disinterest in its gestation, the babe emerged.

We'll briefly review the game's rules, make a strategic assumption, and the solution method for the commercially offered game before diving in to a discovered second quest: asymptotic behavior of infinite dice. We'll note surprising or interesting findings in **bold**.

1.1 The Game

With no physical access to the game set, we search the web for the rules. From what I can only assume is/was the front page of boardgamegeek.com, some intrepid swagman has posted these b.a.d.g. rules [sic]:

1. *Roll all the dice*
2. *Determine if you have bitches a "bitch" is the highest number on any dice usually this is the six but there are a few exceptions the 8 and 10 and 12-sided die as well as the custom*
3. *Remove dice you want to set aside your highest dice to get the best past possible score. You must set aside at least one die per roll you can set aside as many dice as you want.*
4. *Reroll the remaining dice and keep setting them aside until you're out of dice.*
5. *Score add up all the dice The person with the highest score wins.*

1.2 Translation

I'll take it from here, TrevDogg. To formalize this more clearly, a solitaire-style perfect game of b. a. d. g. follows this pattern:

1. Begin with 12 standard dice of six sides, along with one eight-, one ten-, and one twelve-sided die.
2. On each turn, roll all remaining dice in play.
3. If you're looking to play a perfect game, this means that if there are no dice with a maximum side showing, you have lost ("busted").
4. Set aside some number of dice which have achieved their maximum roll.
5. Continue until no dice remain (you have a perfect game) or you have busted out.

The question *"how likely am I to get a perfect score?"* therefore relies on the *strategy* one employs in picking up dice.

1.3 The Strategy Assumption

Proposition 1.1 (Smoke 'em if you got 'em)**.** *The best strategy to achieve a perfect game is to always pick up as many dice with maximum face value as the turn allows.*

Upon reflection, this proposition is:

1. Simple

2. Obvious
3. The strategy any reasonable person would employ
4. Assumed throughout the rest of the paper
5. **Mathematically false**

Why mathematically false? This is only true in some extreme cases - see **The Smoking Gun TODO unbold this** section for details. However, for the discretely sized dice of 6, 8, 10, and 12 sides, we'll assume having more points in the bank and fewer dice to roll is better than the inverse; intuitively, if you set aside a die with a max roll, that die will *again* have to hit a max roll to achieve your perfect game. Why not just bank it?

In the next section, we assume this simple, deterministic strategy in calculating the expected value of any state reachable from the standard b. a. d. g. starting configuration.

1.4 Solution Method for the Commercial Game

These definitions will help us write a program to get our solution for the commercial game:

- The "state" $\vec{S} := [a, b, c, d]$ is how many dice of size 6, 8, 10, and 12 remain, respectively. We start at $\vec{S} = [12, 1, 1, 1]$. A perfect game ends at $\vec{S} = \vec{0}$. A failure to roll any max roll ends the game in an unlisted 'losing' state $\varnothing$.
- The "probability vector" $\vec{p} := [1/6, 1/8, 1/10, 1/12]$ is the chance of getting a max roll from each of the dice sizes, again here assumed to be 6, 8, 10, 12.
- With a win defined as value 1 and a loss as value 0, the "expected value" of being in state $\vec{s}$ is defined as $F(\vec{s})$. Since there is only a single, deterministic strategy, the expected value is then equivalent to the probability of ending up in winning state $\vec{0}$ from state $\vec{s}$.
- Therefore, $F(\vec{0}) = F([0, 0, 0, 0]) = 1$, and $F(\varnothing) = 0$.

First, let's limit ourselves to a game (or state) with only a single flavor of dice, say d-sided, collapsing state vector $\vec{s}$ to a single integer r (r dice of size d left). Refer to Fig. 1 for a visualization of the states starting at r dice left of size d, with $p := 1/d, q := 1 - p$.

Lemma 1.2 (EV of r d-sided dice)**.** *The chance of a completing perfect game from a state of r standard dice of d sides each, with definitions $p := \frac{1}{d}, P_{single}(r, k, p) := \binom{r}{k} p^k (1-p)^{r-k}$, and $F(0) = 1$ is defined by the recurrence $F(k) = \sum_{k=1}^{r} P_{single}(r, k, p) F(r - k)$.*

Proof: This a clear application of the binomial theorem. The chance of transitioning to the state with $r - k$ remaining dice from that with r is the chance of rolling exactly k max

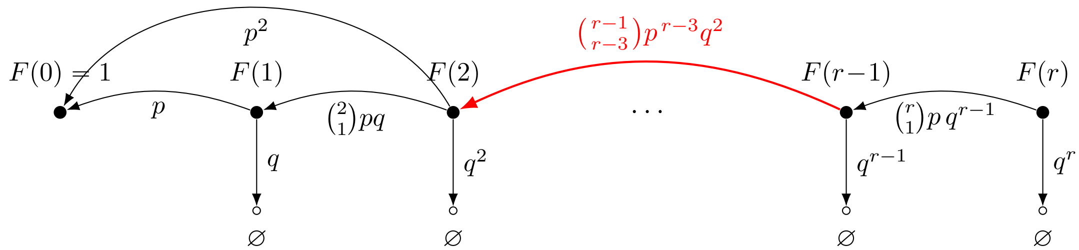

Figure 1: Transition graph of the dice game showing expected values $F(t)$ and transition probabilities.

rolls and $r-k$ non-max rolls among your d-sided dice, or $P_{single}(r,k,p)$. The expected value of being in this new state is defined as $F(r-k)$. The value of this state transition possibility is $P_{single}(r,k,p)F(r-k)$. The total expected value of being in state r is the sum of the values of all possible state transitions. Note that $k=0$ implies a roll with no maxes, and thus a loss (value 0), and $F(0)=1$ since no dice remaining (outside of a loss case) means a victory, or unit value of expectation.

Extending this to dice of different sizes is clear:

Lemma 1.3 (State Transition lemma)**.** *$P_{transition}(\vec{s_1},\vec{s_2},\vec{p})$, the chance of transitioning from state $\vec{s_1}=[s_{11},s_{12}\dots s_{1z}]$ to state $\vec{s_2}=[s_{21},s_{22}\dots s_{2z}]$ with probability vector $\vec{p}=[p_1,p_2\dots p_z]$ is equal to $\prod_{i=0}^{z}P_{single}(\vec{s_1}_i,\vec{s_2}_i,\vec{p}_i)$.*

Proof: Divide the turn's roll into z independent subrolls, each rolling all s_{1i} dice, $1\leqslant i\leqslant z$, each with p_i chance of hitting a max. The outcomes of these z subrolls are all mutually independent, so $P_{transition}$ can be expressed as the product of P_{single} quantities.

1.5 Simple recursive algorithm

With these easy proofs and definitions in hand, the algorithm follows readily - build up each state from $F(\vec{0})$ to $F(\vec{S})$ by applying the State Transition Lemma and previously computed values of F in Algorithm 1. Notes:

- Steps 1 - 5 are definitions from the lemmas above.
- Step 6 creates a set of all states reachable from state $\vec{s}$, *including $\vec{s}$ itself.*
- Step 7 creates an ordered list of states $s_1, s_2, s_3\dots$, like $[0,0,0,0],[0,0,1,0],[0,1,0,0],[0,1,1,0]\dots.$ where, when s_i, is encountered in the list, all reachable states have occurred previously in the list.
- Step 9 calculates $F(\vec{s})$ from the values of previously calculated substates and the use

of Lemma 1.3.

Algorithm 1 Dynamic program for $F(\vec{S})$

$\vec{S} \leftarrow [12, 1, 1, 1]$
$\vec{p} \leftarrow \left[\frac{1}{6}, \frac{1}{8}, \frac{1}{10}, \frac{1}{12}\right]$
$P_{\text{single}}(r, k, p) := \binom{r}{k} p^k (1-p)^{r-k}$
$P_{\text{transition}}(\vec{s_1}, \vec{s_2}, \vec{p}) := \prod_{i=0}^{\dim(\vec{s})} P_{\text{single}}(\vec{s_1}_i, \vec{s_2}_i, \vec{p}_i)$
$F(\vec{0}) \leftarrow 1$
$\text{substates}(\vec{s}) := \bigtimes_{i=0}^{dim(s)} [0, s_i]$
$\text{states} \leftarrow \text{topological\_sort}\Big(\text{substates}(\vec{S}) \setminus \{[0,0,0,0]\}\Big)$
for $\vec{s} \in$ states **do**
    $F(\vec{s}) \leftarrow \sum_{\vec{t} \in (\text{substates}(\vec{s}) \setminus \{\vec{s}\})} P_{\text{transition}}(\vec{s}, \vec{t}) \cdot F(\vec{t})$
end for

1.6 Results for Commercial Game

The chance of winning[1] $F(\vec{s})$ for every state $\vec{s}$, with $\vec{s}$ ranging from $\vec{0}$ to $\vec{S} = [12, 1, 1, 1]$ is listed in Figures 2 and 3. These are computed with the simple python program at https://github.com/fettermania/mathnotes/blob/main/bdice2/bdice2.py.

Note that these are *four-dimensional* results, with each dimension i corresponding to the count of dice s_i in each state. Since the number of 8-, 10-, and 12-sided dice is either zero or one, we've collapsed these into four 12x2 tables, where d_k notes the number of dice of size k.

Our main *initial* result is in the bottom-right corner of Fig. 3.

> **What The Creator Asked For**: The chance of getting a perfect run the commercial game using the clear strategy is about 0.004413, or a little more than 4.4%.

1. State 0.004312: When you roll exactly 2 sixes (probability = 21%) - Naive: Take both sixes ? [8,1,1,1] - Optimal: Take only 1 six ? 0.004284
2. State 0.004366: When you roll exactly 2 sixes (probability = 21%) - Naive: Take both sixes ? 0.004284 - Optimal: Take only 1 six ? 0.004312
3. State 0.004446: When you roll exactly 2 sixes (probability = 21%) - Naive: Take both sixes ? 0.004312 - Optimal: Take only 1 six ? 0.004366

[1] Alternatively, expected value of being in state $F(\vec{s})$, with $F(\vec{0}) = 1, F(\varnothing) = 0$

$d_6 \backslash d_8$	0	1
0	1	0.125
1	0.1667	0.05642
2	0.07407	0.03244
3	0.04192	0.02206
4	0.02809	0.01679
5	0.02107	0.01378
6	0.01708	0.01195
7	0.01463	0.01076
8	0.01303	0.009966
9	0.01195	0.009423
10	0.01119	0.009044
11	0.01065	0.008779
12	0.01025	0.008591

$d_6 \backslash d_8$	0	1
0	0.1	0.03469
1	0.04556	0.02046
2	0.02644	0.01425
3	0.01815	0.01109
4	0.01392	0.009299
5	0.01152	0.008218
6	0.01006	0.007537
7	0.009124	0.0071
8	0.008506	0.006818
9	0.00809	0.006638
10	0.007809	0.006529
11	0.007618	0.006468
12	0.007491	0.006441

Figure 2: Expected values for $(d_{10}, d_{12}) = (0, 0)$ (left) and $(1, 0)$ (right).

Observations from Figs. 2 and 3:

- In almost all cases, the more dice remain, the lower the expected value of winning is.
- However, we do note some non-monotonicity: there's are two surprising "bends" at $[11, 0, 1, 1]$ and $[8, 1, 1, 1]$.
- At all states reachable from $[12, 0, 1, 1]$, the best strategy is confirmed: pick up as many max-rolled dice as possible. Deviating from this strategy produces an F-value smaller than those in the table. *Note:* The left-column "bend" at $[11, 0, 1, 1]$ does not affect this; you have no alternative from [12,0,1,1] than to pick up if [11,0,1,1] is your new roll state.
- At all states reachable from $[9, 1, 1, 1]$, the simple strategy remains best.
- However, if you dice s
- We can confirm that holding d_6 steady, it's always better to have zero than one of each of the 8, 10, and 12-sided dice.
- However, if $d_{10} = 1$ and $d_{12} = 1$ we notice that it's slightly *better* to have 12 d_6 dice to roll than 11 or 10.

$d_6 \backslash d_8$	0	1
0	0.08333	0.02908
1	0.03819	0.01726
2	0.02231	0.01209
3	0.0154	0.009464
4	0.01188	0.00798
5	0.009888	0.007088
6	0.008677	0.006532
7	0.007907	0.006181
8	0.007404	0.00596
9	0.007071	0.005825
10	0.006851	0.00575
11	0.006707	0.005715
12	0.006618	0.005709

$d_6 \backslash d_8$	0	1
0	0.02347	0.01087
1	0.01406	0.007809
2	0.009946	0.006252
3	0.007849	0.005385
4	0.006672	0.004879
5	0.005971	0.00458
6	0.005542	0.004408
7	0.005278	0.004318
8	0.00512	0.004283
9	0.005033	0.004284
10	0.004994	0.004312
11	0.004988	0.004356
12	0.005005	0.004413

Figure 3: Expected values for $(d_{10}, d_{12}) = (0, 1)$ (left) and $(1, 1)$ (right).

1.7 Re-examining The optimal strategy

From examining the table it becomes clear that there are expected values for states $\vec{s}, \vec{t}$ such that $\vec{t}$ precedes $\vec{s}$ yet $F(\vec{t}) < F(\vec{s})$.

This means that for state, say, $\vec{s} + e_1$, that even when rolling enough maxes to get to $\vec{t}$, picking up fewer dice to get to will beat the Smoke 'em if you Got em strategy.

This means we need to refine our formula to

$selectable\_states(\vec{s_1}, \vec{s_2}) := substates(\vec{s_1} - \vec{s_2}) + \vec{s_2}$
$\{\vec{s_1}\}$.

$F(\vec{s}) \leftarrow \sum_{\vec{t} \in (\text{substates}(\vec{s}) \setminus \{\vec{s}\})} P_{\text{transition}}(\vec{s}, \vec{t}) \cdot max_{\vec{u} \in selectable\_states(\vec{s_1}, \vec{s_2})} F(\vec{u})$

Because F is monotonic up to these inflection points, these two functions will be equal on earlier states.

So, we need to recursively reevaluate F on states "after" $[12, 0, 0, 1]$ and $[8, 1, 1, 1]$ knowing that $F* = F$ before then.

So it turns out Mr. Cox's students are correct - you may indeed want to "save" future six rolls if your rarer "onesies" remain. The difference on the commercial game begins at 9,1,1,1 and can increase your odds to xxx.

It's nice to see the game is complex and non monotonic even in this constrained scenario. There are likely higher configurations yielding other critical points in the space e, especially

$d_6 \backslash d_8$	0	1
0	0.02347	0.01087
1	0.01406	0.007809
2	0.009946	0.006252
3	0.007849	0.005385
4	0.006672	0.004879
5	0.005971	0.00458
6	0.005542	0.004408
7	0.005278	0.004318
8	0.00512	0.004283
9	0.005033	0.004284
10	0.004994	0.004312 → 0.004312
11	0.004988	0.004356 → 0.004366
12	0.005005	0.004413 → 0.004446

Figure 4: Expected values for $(d_{10}, d_{12}) = (0, 1)$ (left) and $(1, 1)$ (right).

as more dimensions (dice types) are added.

But can we say nothing more definitive than that? Looking at the first table (fig 2), with only one type of die, it looks like a higher dice count both decreases our chance of winning, but may also do so asymptotically.

Let's run $\vec{0} + 100e_1$ and see.

Todo

It certainly looks like there is an asymptote to which f(s) converges. This will be the thrust of the rest of this paper: examining the infinite game of F(s) with one type of die.

Lemma 1.4 (Easy Cases Lemma)**.** *If* $p = 0$*, F(r) = 0. If* $p = 1, F(r) = 1$*. If* $s = 2$*, for any* $r > 0, F(r) = \frac{1}{2}$*.*

Proof: Though we won't be flipping infinite ($p = 0$) or 1-sided ($p = 1$), dice, these algebraic identities are obvious (can't succeed, can't fail, respectively) and will be useful later.

Base case for s=2: $F(1) = \frac{1}{2}$ since a perfect game occurs if and only if the coin flips 2 instead of 1.

Inductive case for s=2: Assuming $F(k) = \frac{1}{2}$ for all $0 < k < r$, and $F(0) = 1$:

$$\sum_{k=1}^{r-1} \binom{r}{k} p^k q^{r-k} F(r-k) + \binom{r}{r} p^r q^0 F(0) = \tag{1}$$

$$(2^r - 2)\frac{1}{2^r}(\frac{1}{2}) + \frac{1}{2^r}2(\frac{1}{2}) = \tag{2}$$

$$(2^r)\frac{1}{2^r}(\frac{1}{2}) = \frac{1}{2} \tag{3}$$

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- Extremely Lazy Lower bound (closed form)
- Wrong attempt: $F(t) < F(t-1)$?
- F(n) exhibits provably asymptotic behavior

Theorem 1.5. *For an unbounded set of $r > 0$ standard s-sided dice, $s \geqslant 2$, the probability $F(r)$ of getting a perfect game has a lower bound of* $\exp(-\frac{q}{(1-q)^2})$*, if we define $q := 1 - \frac{1}{s}$.*

Proof:

- First, while at stage k, where k dice remain, the probabilty $P_s(k)$ of continuing to some stage $j < k$ is clearly $1 - q^k$.
- The probability $F(k)$ of passing all such stages is greater than or equal to $P^* = \prod_{i=1}^{k} P_s(k)$, since a successful traversal may involve skipping one or more stages, and P^* assumes we no stages.
- Therefore, $F(k) \geqslant \prod_{i=1}^{k} P_s(k) = \prod_{i=1}^{k}(1 - q^k)$.
- This means $\log(F(k)) \geqslant \sum_{i=1}^{k} \log(P_s(k)) = \sum_{i=1}^{k} \log(1 - q^k)$.

This $F(k)$ is apparently defined as the **q-Pochhammer symbol** $(q;q)_\infty = \prod_{i=1}^{k}(1 - q^k)$, with known convergence properties. I'll instead attempt to tackle convergence without the aid of these theorems.

Often, $\log(1-z)$ is approximated by $-z$ near zero, but we are most concerned with z near one, since $1-z$ approaches 0, and $\log(1-z)$ explodes. Therefore, let's look at the expansion of $\log(1-z)$:

- Identity 1: $f(z) = -\frac{1}{1-z} = -1 - z - z^2 - \dots$
- Identity 2: $\int f(z) = \log(1-z) = -z - \frac{z^2}{2} - \frac{z^3}{3} - \dots$

So when summing $\sum_{i=1}^{k} \log(1-q^k)$, we can instead sum

$$\log(F(k)) = \log(1-q^1) + \log(1-q^2) + \log(1-q^3) + \dots \tag{4}$$

$$\log(F(k)) = (-q - \frac{q^2}{2} - \frac{q^3}{3} + \dots) + (-q^2 - \frac{q^4}{2} - \frac{q^6}{3} + \dots) + (-q^3 - \frac{q^6}{2} - \frac{q^9}{3} + \dots) + \dots \tag{5}$$

$$> (-q - q^2 - q^3 + \dots) + (-q^2 - q^4 - q^6 + \dots) + (-q^3 - q^6 - q^9 + \dots) + \dots \tag{6}$$

$$= \frac{-q}{1-q} + \frac{-q^2}{1-q^2} + \frac{-q^3}{1-q^3} + \dots \tag{7}$$

$$> \frac{-q}{1-q} + \frac{-q^2}{1-q} + \frac{-q^3}{1-q} + \dots \tag{8}$$

$$= \frac{1}{1-q}[-q - q^2 - q^3 - \dots] \tag{9}$$

$$= \frac{-q}{(1-q)^2} \tag{10}$$

- (2) follows from (1) by substitution using Identity1 above.
- (3) follows from (2) since the denominators of (2) are larger.
- (4) follows from (3) when dividing each sequence by $-q^k$ and applying identity 2.
- (5) follows from (4) since the denominators of (4) are larger.
- (6) and (7) follow from (5) by dividing out $\frac{1}{1-q}$ and applying Identity 1.

Since $\log(F(k)) > \frac{-q}{(1-q)^2}$, $F(k) > \exp(\frac{-q}{(1-q)^2})$.

This is a thoroughly unsatisfying lower bound, as we know $F(2) = \frac{1}{2}$.

s	p	q	bound
2	$\frac{1}{2}$	$\frac{1}{2}$	$\exp(-2) \approx .135$
3	$\frac{1}{3}$	$\frac{2}{3}$	$\exp(-6) \approx .00248$
4	$\frac{1}{4}$	$\frac{3}{4}$	$\exp(-12) \approx 6.14 \times 10^{-6}$
10	$\frac{1}{10}$	$\frac{9}{10}$	$\exp(-90) \approx 8.19 \times 10^{-40}$

Table 1: Very loose lower bounds on $F(k = \infty)$

Definition 1.6 ($(q;q)_\infty$: The q-pochhammer symbol)**.** *The q-pochhammer symbol* $(q;q)_\infty = \prod_{k=1}^{\infty}(1-q^k)$ *is a known convergent quantity.*

q	$(q;q)_\infty$	$\exp\left(-\frac{q}{(1-q)^2}\right)$
1/2	2.8879×10^{-1}	1.3534×10^{-1}
2/3	6.9272×10^{-2}	2.4788×10^{-3}
3/4	1.5545×10^{-2}	6.1442×10^{-6}
4/5	3.3680×10^{-3}	2.0612×10^{-9}
5/6	7.1400×10^{-4}	9.3576×10^{-14}
6/7	1.4913×10^{-4}	5.7495×10^{-19}
7/8	3.0815×10^{-5}	4.7809×10^{-25}
8/9	6.3155×10^{-6}	5.3802×10^{-32}
9/10	1.2861×10^{-6}	8.1940×10^{-40}

However, this does provide a positive lower bound, and brings up this interesting fact: **Though increasing the number of sides of the dice can drive F(k) to zero, increasing k to ∞ will never set F(k) to zero**.

If we can couple this with the proof of the next theorem, we can state that $F(k)$ has an asymptotic limit as $k \to \infty$ given some q.

Calculating by hand:

- $F(0) = 1$
- $F(1) = p$
- $F(2) = p^2(1 + 2q) = p^2(3 - 2p)$
- $F(3) = p^3(13 - 27p + 21p^2 - 6p^3)$
- $F(4) = p^4(75 - 316p + 606p^2 - 664p^3 + 432p^4 - 156p^5 + 24p^6)$
- $F(1) - F(0) = p - 1 = -(1 - p)$
- $F(2) - F(1) = p^2(3 - 2p) - p = (-1)p(1 - 2p)(1 - p)$

- $F(3) - F(2) = (-1)3p^2(1-p)^3(1-2p)$
- $F(4) - F(3) = -p^3(1-2p)(p-1)^2(12p^4 - 48p^3 + 72p^2 - 50p + 13)$

Theorem 1.7 (Incorrect Blind Alley: $F(n) < F(n-1)$)**.** *If $p < \frac{1}{2}, F(n-1) > F(n)$ for $n > 1$.*

- TODO: Show calculation of various functions directly.
- Show calculation of $p = \frac{1}{2} - \epsilon, q = \frac{1}{2} - \epsilon$ with $p = 0.495$.
- Show interesting observations: growth of polynomial
- $R(t)$ is starting at position t, throwing $t-1$ dice.
- $F(t) = p(1 + q^{t-1})F(t-1) + qR(t) = pF(t-1) + pq^{t-1}F(t-1) + qR(t)$
- If $qR(t) < qF(t-1) - pq^{t-1}F(t-1)$, same as $F(t-1) - R(t) > pq^{t-2}F(t-1)$, then $F(t) < F(t-1)$.
- In the special case $q = p = \frac{1}{2}$, then $F(t-1) - R(t) = p^{t-1}(1) - p^{t-1} \cdot p +$ other zero factors $= p^t = pq^{t-2} \cdot \frac{1}{2}$.
- So we're satisfied since $F(t) = \frac{1}{2}$ for $t \geqslant 1$.
- If $p \leqslant \frac{1}{3}, F(t-1) - R(t) = [p^{t-1}F(0) - p^{t-1}F(1) +$ other positive factors $] = p^{t-1}q + \epsilon$.

$$t = 3, F(3) - R(4) > pq^{4-1}F(4-1) \tag{11}$$
$$p^3F(0) - p^3F(1) \tag{12}$$
$$\binom{3}{2}p^2qF(1) - \binom{3}{2}p^2qF(2) \tag{13}$$
$$\binom{3}{1}p^2qF(2) - \binom{3}{2}p^2qF(3) \tag{14}$$
$$\tag{15}$$

2 Interesting Findings

- Calculating the asymptotic behavior of a combination of s-sided dice and t-sided dice isn't particularly interesting; if $s > t$, this less than $F_{1/t}(\infty)$ and more than $F_{1/s}(\infty)$.
- no closed form - triangle exponent
- non monotone commercial game
- non monotone single game

- pochhammer bounds, ultimate convergence
- can be made zero in p, not in n
- note: different than best strategy (since others have knowledge)